

Data Collection and Corpus-Construction Guide

1. Purpose and scope

This guide tells the research team how to collect data and build the two text corpora that feed the study's computational-qualitative analysis. Corpus A is the authoritative institutional and governmental archive, read for the institutional and informational hurdles. Corpus B is the application-programming-interface-accessible community-discourse corpus, read for the lived internal and relational hurdles. The two are analysed with transformer-based topic modelling and interpreted abductively against a four-locus hurdle scaffold, namely internal capability and resources, relational power and dependence, institutional voids and legitimacy, and informational verifiability.

The work is a secondary qualitative inquiry on publicly available and institutional data, so it follows a procedural secondary-qualitative methodology designed to increase rigour, mitigate the known pitfalls of reusing data produced for other purposes, and document every decision (Cheong et al., 2023). The commodity scope is held bounded and is not expanded to a census. Spices and marine products are the core pair; rice and the deforestation-regime commodities form the comparison ring; processed and value-added food is added to the ring as a third verification logic, because it documents proof differently again, through process-and-facility certification and multi-ingredient traceability rather than single-commodity residue testing or catch-and-aquaculture control.

Commodity element	Role	How proof is documented	What it isolates
Spices	Core	Pesticide, residue and contaminant testing on a single commodity; RASFF anonymised	Informational and institutional hurdles under chemical-contaminant scrutiny
Marine products	Core	Catch legality, aquaculture residue, pre-harvest and pre-export testing; FDA firm-named	Relational and informational hurdles under a catch-and-aquaculture regime
Processed and value-added food	Ring, third logic	Process-and-facility certification (HACCP, BRCGS, IFS, FSSC 22000, SQF), multi-ingredient traceability	Hurdles where identity is most pooled, the extreme case of proof detachment
Rice	Ring	Lower-intensity regime, baseline	A setting where the proof demand is weaker
Deforestation-regime commodities	Ring	A regime that does not cover the core	Separates regime-driven hurdles from structurally produced ones

2. How this corpus answers the central question

The central question is what hurdles keep an Indian agri-food MSME from becoming a recognised sustainable supplier, how they distribute across the four loci, how they interlock, how they are framed, and how they move. The corpus answers it through four design features, namely locus coverage enforced by the closing audit, controlled contrast across the three verification logics, a question-to-operation mapping in which each sub-question has a named analytical operation, and triangulation of the official record against the lived discourse with the comparison ring supplying negative cases. The answerability boundary is stated honestly, namely that the corpus answers the hurdle-landscape question at population and discourse level, not a firm-level causal question, which would require primary data.

Methodology Protocol

1. Research questions

RQ1. What hurdles prevent recognition, and how are they distributed across the four loci, namely internal capability, relational power, institutional voids and informational verifiability?

RQ2. How do these hurdles interlock, that is, which co-occur and condition one another to form the configurations a firm faces?

RQ3. How do different actors, namely regulators, buyers, certifiers, intermediaries and firms, frame the same hurdles, and where do those framings diverge?

RQ4. How does the hurdle landscape move over time around regulatory announcements, postponements and softenings?

2. Design and logic

The study is a computational-qualitative analysis of secondary, unobtrusive text, abductive in that it brings a four-locus scaffold to the data while remaining open to emergent categories. Computation makes qualitative analysis possible at scale while human interpretation remains the source of meaning (Evans, 2014), and the community corpus is treated as a netnographic object, since computational text mining without immersion is not netnography (Rynarzewska and Giunipero, 2024). The document is organised so that a shared foundation, Section 3, is built once, and each research question is then answered by its own procedure, Sections 4 to 7, that draws on that foundation.

3. Methodological foundations and rigour stance

Because this is secondary qualitative research on data produced for other purposes, rigour is not assumed from the size of the corpus but is built in procedurally. Four commitments govern the build.

1. A procedural secondary-qualitative methodology. The build follows a staged procedure with an explicit data-quality assessment that screens each source on the quality of its context and the

quality of its content before it enters the corpus, so that unfit data is filtered out rather than silently included (Cheong et al., 2023).

2. Context sensitivity. Secondary data must be read in the light of why, by whom and for whom it was originally produced, so every source is documented with its production context, which is the condition under which reuse is defensible (Cheong et al., 2023).
3. Open-science transparency, with honest limits. The protocol, the registry and the decision log are prepared so that the build is auditable and as far as possible reproducible, while recognising that full sharing of qualitative and community data is constrained by ethics and platform terms and cannot always be achieved (Lamb et al., 2024).
4. Ethics for qualitative big data. Community discourse is treated under a big-data research ethic that does not assume scale removes the duty of care, and that protects contextual integrity and identifiable persons (Stegenga et al., 2023), operationalised through the de-identification protocol in section 11.

Trustworthiness is pursued against the four established criteria, namely credibility, transferability, dependability and confirmability, each mapped to a concrete practice in section 13.

4. Operating principles

1. Reproducibility. Every document and every query is logged with a source, a URL or query string, and a retrieval date.
2. Provenance and context. Every source carries a note on its institutional origin, its purpose of production and its known biases.
3. Fitness before inclusion. No source enters the corpus until it passes the data-quality assessment in section 6.
4. Four-locus and three-logic coverage. The corpus must populate every hurdle locus and every verification logic, checked at the closing audit.
5. Saturation, not exhaustion. The corpus is grown until new sources stop adding new content, and the build states what was not reached.
6. Ethics and de-identification first for community data, under the protocol in section 11, cleared by the institution before collection.

5. The master registry

All collections are logged in a single master registry, one row per document or per discrete data pull. The registry is the spine of the build, the source of the coverage statement, and the core of the audit trail.

Column	Content
doc_id	Unique identifier, for example A-VAF-0012
corpus / tier	A or B; source tier from sections 7 and 8

Column	Content
source_name / instrument_or_body	Issuing body or platform; named instrument where relevant
title / url_or_query	Document title; direct URL or exact query string
retrieval_date / format / language	ISO date; pdf, html, csv, json, api; source language
production_context	Who produced it, why, for whom, when (secondary-data context note)
dqa_context_score / dqa_content_score	Data-quality assessment results from section 6
full_text_available	yes, abstract, metadata-only
locus_tag / verification_logic	Hurdle locus; proof regime
deident_status	not-required, pending, completed (Corpus B)
access_status / notes	retrieved, paywalled, restricted, archived; provenance and bias note

6. Data-quality assessment, the inclusion filter

Each candidate source is screened on two axes before inclusion, following the context-and-content quality logic for secondary online data (Cheong et al., 2023). A source that fails the screen is excluded and the exclusion is recorded, so that the corpus is filtered by an explicit standard rather than by convenience. Score each dimension as adequate, marginal or inadequate, and admit a source only if no dimension is inadequate.

6.1 Context quality

Dimension	Question
Authority	Is the issuing body authoritative for this matter
Purpose of production	Why was the document produced, and does that purpose bias it
Audience	For whom was it written, and does that shape what it records
Provenance and integrity	Is the source the original or a faithful copy, and is it complete
Recency and currency	Is it current for the regime or period in question
Accessibility and stability	Can it be retrieved again at a stable address

6.2 Content quality

Dimension	Question
Relevance	Does it speak to a hurdle, an actor, a commodity or a regime in scope

Dimension	Question
Richness	Does it carry enough substantive text to be informative, not a stub
Specificity	Is the content concrete rather than generic boilerplate
Credibility and consistency	Is it internally consistent and consistent with corroborating sources
Locus signal	Can the content be mapped to at least one hurdle locus
Verification-logic tag	Can it be assigned to a proof regime for the contrast

7. Building Corpus A, the institutional archive

Corpus A is built tier by tier. Capture full text where available, record the URL and retrieval date and the production-context note, run the data-quality assessment, and place anything inaccessible in the exceptions file.

7.1 India study-area-specific, the niche official core

Source and document	Direct link	Logic / locus
APEDA TraceNet (NPOP organic), HortiNet, and the commodity tracing systems	https://apeda.gov.in ; https://agriexchange.apeda.gov.in	all / informational
APEDA processed food and value-added export categories	via APEDA portal	value-added / institutional
Spices Board ESS and CRES; Salmonella, Japan and EU health-certificate circulars	https://indianspices.org.in/ESS/	spices / institutional
MPEDA exporters portal and e-SANTA	https://mpeda.gov.in/exporters/	marine / relational
Export Inspection Council, notified products and EIA labs	https://www.eicindia.gov.in/WebApp1/	all / institutional
FSSAI processing and manufacturing licences, standards, alerts	https://www.fssai.gov.in	value-added / institutional
India Trade Portal; DGFT IEC register and Foreign Trade Policy	https://www.indiantradeportal.in ; https://www.dgft.gov.in	all / institutional
Open Government Data and DGCI and S trade statistics	https://www.data.gov.in	all / informational
NITI Aayog reports; Ministry of MSME Udyam and Zero Defect Zero Effect	https://www.niti.gov.in ; https://zed.msme.gov.in	all / internal
Corporate Social Responsibility portal, Section 135 filings	https://www.csr.gov.in	all / institutional

7.2 Snowball seed list from the DG SANTE India audit (instruments and bodies)

EU instruments: Regulation (EU) 2017/625 (Official Controls); Decision 2011/163/EU; Regulations (EU) 2021/404 and 2021/405; Decision 2010/381/EU (amended by 2012/690/EU and 2016/1774); Directives 96/22/EC and 96/23/EC; Decision 97/747/EC; Regulation (EU) 2019/2090; Regulation (EC) 470/2009 and Regulation (EU) 37/2010; Regulation (EU) 2021/808 (replacing Decision 2002/657/EC); Regulation (EC) 396/2005; Regulations (EC) 1881/2006 and 333/2007; Regulation (EU) 2019/1871; Regulation (EU) 2019/6.

Indian instruments: Export (Quality Control and Inspection) Act 1963; Notification S.O. 497(E) of 10 March 2011; S.O. 4031(E) and 4032(E) of 9 November 2020; S.O. 1377(E) and 1378(E) of 30 December 2002; Order S.O. 722(E) of 10 July 2002; Order S.O. 1227(E) of 23 October 2003; Drugs and Cosmetics Act 1940 and Rules 1945; S.O. 730(E) of 21 August 1998; Kerala Fish Seed Act 2014; Kerala Inland Fisheries and Aquaculture Act 2010; Andhra Pradesh State Aquaculture Development Authority Act 2020.

Bodies: European Commission DG SANTE; Export Inspection Council and the Export Inspection Agencies (Delhi, Mumbai, Kochi, Kolkata, Chennai); MPEDA and its NaCSA arm; Coastal Aquaculture Authority; State Fishery Authorities; Central Drugs Standard Control Organisation; State Drug Licensing Authorities; Department of Animal Husbandry; FSSAI; Central Institute of Fisheries Technology; National Accreditation Board for Calibration and Testing Laboratories; Seafood Exporters Association of India; the EU Reference Laboratories; and the International Laboratory Accreditation Cooperation.

7.3 European Union, United States and multilateral

Source and document	Direct link	Logic / locus
EUR-Lex: IUU Regulation, MRL Reg 396/2005, CSDDD, CSRD, composite and additive rules	https://eur-lex.europa.eu	all / institutional
Reg (EU) 2019/1793, increased controls on Indian spices and certain processed foods	https://eur-lex.europa.eu/eli/reg_impl/2019/1793/oj	spices, value-added / institutional
RASFF Window (product-and-origin level, not firm-named)	https://webgate.ec.europa.eu/rasff-window/screen/search	all / informational
DG SANTE audit reports for India	https://food.ec.europa.eu/audits-analysis	marine, value-added / institutional

Source and document	Direct link	Logic / locus
EUMOFA market reports; EFSA residue reports	https://www.eumofa.eu ; https://www.efsa.europa.eu	marine, spices / informational
FDA import refusals (firm-named) and import alerts	https://www.accessdata.fda.gov/scripts/importrefusals/	all / informational
FDA Foreign Supplier Verification Program and facility registration (FSMA)	https://www.fda.gov/food/importing-food-products-united-states	value-added / institutional
ITC Standards Map; WTO ePing; OECD-FAO Guidance; UN Comtrade; UNCTAD TRAINS; FAOSTAT; ILO NORMLEX	see registers and multilateral links	all / institutional

7.4 Standards indexed in ITC Standards Map, relevant to scope

Confirm each record in the live tool at <https://www.intracen.org/resources/tools/standards-map>. Marine and aquaculture: Aquaculture Stewardship Council, Best Aquaculture Practices, Marine Stewardship Council, GLOBALG.A.P. Aquaculture, Naturland, Friend of the Sea. Spices and produce: Rainforest Alliance, Fairtrade International, EU Organic, USDA NOP, India NPOP, GLOBALG.A.P. Integrated Farm Assurance, Sustainable Spices Initiative, Fair for Life. Cross-cutting social and ethical: SA8000, amfori BSCI, Sedex SMETA, ETI Base Code.

7.5 Certification, accreditation and process-based registers

The first group is the certified-operator layer, the signal. The second is the accreditation layer, the warrant behind the signal. The process-based schemes carry the value-added verification logic.

Register	Direct link	Layer
Best Aquaculture Practices certified facilities	https://bap.globalseafood.org	signal
Aquaculture Stewardship Council; Marine Stewardship Council	https://asc-aqua.org/find-a-farm/ ; https://www.msc.org	signal
GLOBALG.A.P. certified producers database	https://database.globalgap.org	signal
USDA Organic Integrity Database; FLOCERT Fairtrade search	https://organic.ams.usda.gov/integrity/ ; https://www.flocert.net	signal

Register	Direct link	Layer
SA8000 certified organisations	https://sa-intl.org/sa8000-search/	signal
BRCGS Directory; FSSC 22000 certified organisations	https://directory.brcgs.com ; https://www.fssc.com/certified-organizations/	signal (value-added)
IFS Food database (partly restricted); SQF certified suppliers	https://www.ifs-certification.com ; https://www.sqfi.com	signal (value-added)
SAAS accredited certification bodies; ISEAL community members; GFSI recognised schemes	https://www.saasaccreditation.org ; https://www.isealalliance.org ; https://mygfsi.com	warrant

8. Building Corpus B, the community discourse

Corpus B is built platform by platform through application-programming interfaces, beginning with the most usable, namely Reddit, YouTube and GDELT, then the paid X interface, then trade media within terms of service. Record the exact query string, the date range and the parameters, and run the same data-quality assessment and the de-identification protocol before any text enters analysis.

8.1 News and event databases

The accepted free backbone for media discourse read at scale is GDELT, which draws on open-web news and is run from academia, while Factiva, LexisNexis and Event Registry provide verifiable full text at a cost, and Media Cloud serves open-web media-ecosystem analysis. GDELT carries a known accuracy and redundancy cost, so deduplicate and clean before analysis.

Database	Access	Best for this study	Link
GDELT	Free, API	Large-scale event and tone discovery, dynamics over the regulatory timeline	https://www.gdeltproject.org
Factiva	Subscription	Verifiable full-text trade and business news	https://professional.dowjones.com/factiva/
LexisNexis (Nexis)	Subscription	Full-text, longitudinal, structured metadata	https://www.lexisnexis.com
Event Registry	Subscription / API	Multilingual event clustering, full text	https://eventregistry.org
Media Cloud	Free / open	Open-web media ecosystem, hyperlink structure	https://www.mediacloud.org

8.2 Platforms and practitioner discourse

Platform and route	What it surfaces	Locus
Reddit via the PRAW API	Practitioner sensemaking in trade, India, agri, aquaculture, food-processing subreddits	internal
X via API	EUDR, CSDDD and exporter and NGO discourse	relational
YouTube Data API	APEDA, MPEDA, Spices Board channels, food-processing webinars, comments	internal
Trade media within terms of service	SeafoodSource, Undercurrent News, BusinessLine, food-processing trade press	relational
NGO and certifier discourse	Environmental Justice Foundation, Mighty Earth, Changing Markets	institutional

Source	What it holds	Locus	Access	
Indian export trade bodies: FIEO, Seafood Exporters Association of India, All India Spices Exporters Forum, All India Food Processors' Association	Member position representations on compliance burden	Member circulars, papers, on	relational, institutional	Association websites
LinkedIn industry groups on agri-export, spices, seafood and EUDR or CSDDD compliance	Practitioner discussion of buyer demands and compliance hurdles		internal, relational	LinkedIn, within terms
Facebook and producer-organisation groups	Exporter and farmer-producer sensemaking, as in the netnography SONJO study		internal	Graph API, public groups
Reddit communities on trade, India business and agriculture	Candid practitioner problem-talk		internal	PRAW API

YouTube channels of APEDA, MPEDA and the Spices Board, and agri-export educators, plus comments	Official guidance and audience reaction	internal, institutional	YouTube Data API
X hashtags and handles, for example EUDR and CSDDD, exporter and NGO accounts	Real-time reaction	regulatory relational	X API
Trade media: The Hindu BusinessLine, SeafoodSource, Undercurrent News, Fresh Plaza	Sector reporting on rejections, standards and market access	relational, informational	Site or Factiva
NGO and certifier discourse: EJF, Mighty Earth, Changing Markets, Sustainable Spices Initiative	Advocacy framing of hurdles	institutional	Organisation sites
Industry webinar and podcast transcripts from trade bodies and certifiers	Expert framing of compliance pathways	institutional, informational	Public recordings
Farmer-producer organisation public forums, de-identified	Upstream producer voice	internal	Public groups, ethics-gated

9. Query protocol

Queries cross three dimensions and every string is logged verbatim. The commodity dimension has five elements, namely spices, marine, processed and value-added food, rice, and the deforestation-regime commodities. The regime dimension covers the deforestation regulation, the due-diligence and reporting directives, the import-control regulation, the pesticide and residue regimes, and the food-safety gate including the process-based schemes. The hurdle dimension covers cost, capability, traceability, certification, intermediary, buyer demand, rejection and refusal, with value-added proof terms added, namely HACCP, food-safety management system, facility certification, multi-ingredient traceability, allergen, additive and blend. A query crosses one term from each dimension, and the query-string library is kept as a living appendix so coverage across dimensions and verification logics can be audited.

Criterion	Filter
Search term	("sustainab*" OR "due diligence" OR "traceab*" OR "certification" OR "residue" OR "MRL" OR "rejection" OR "RASFF" OR "EUDR" OR "CSDDD") AND "India" AND atleast3(export) AND ("spice*" OR "seafood" OR "marine" OR "shrimp" OR "rice" OR "processed food")
Language	English, with Hindi and regional-language items flagged for recorded translation
Industry	Food and Beverage, Agriculture, Fisheries and Aquaculture, Wholesale and Retail Trade
Publication type	Newspapers, Industry Trade Press, Newswires and Press Releases, Trade Magazines and Journals, Industry Analyst Reports
Timeline	01 Jan 2015 to collection date, aligned to the regulatory window before and after the import-control, deforestation and due-diligence regimes

Each API platform then gets its own short specification, for example a Reddit row-set naming the subreddits, the query terms, the date range and the result cap, and a GDELT row-set naming the query, the country and theme filters and the window. Every one of these tables is reproducible, and every retrieval it produces is still logged in the master registry and screened by the data-quality assessment.

10. Cleaning and preprocessing

1. Deduplicate within and across sources, recording counts removed, which matters most for GDELT and news pulls.
2. Language-tag and translate non-English text with a recorded model and version, keeping the original alongside the translation.
3. Strip boilerplate while preserving substantive text; segment long documents into coherent units, recording the rule.
4. Write the cleaned corpus to a fixed schema with doc_id, locus_tag and verification_logic preserved, so every topic traces to its source and proof regime.

11. De-identification and ethics protocol

Community discourse is collected and processed only under a documented protocol grounded in the ethics of qualitative big data, which holds that scale does not remove the duty of care and that

contextual integrity and identifiable persons must be protected (Stegenga et al., 2023), and in the secondary-online-data ethical principles of autonomy, equity and diversity (Cheong et al., 2023).

1. Public-versus-private judgment. Classify each platform and exclude private or closed spaces unless explicit permission exists; record the judgment per source.
2. Structured de-identification of narrative data. Apply an empirically grounded de-identification procedure to all Corpus B text before analysis and before any quotation, removing handles, names, locations and other identifying detail while preserving analytic meaning, and record the deident_status in the registry (Campbell et al., 2023).
3. No re-identifiable quotation. Quotations used in outputs are paraphrased or sufficiently generalised that the speaker cannot be re-identified, consistent with the de-identification standard above.
4. Terms of service and interfaces. Prefer official application-programming interfaces over scraping and respect each platform's terms.
5. Ethics clearance and a sharing statement. Prepare an ethics statement describing platforms, the public nature of the data, the de-identification rule, storage and retention, and clear it with the institution before collection; state honestly what can and cannot be shared, since qualitative and community data sharing is constrained (Lamb et al., 2024).

12. Provenance, audit trail and reproducibility

The build is documented so that an independent researcher could follow and, within ethical limits, reproduce it. Three artefacts constitute the audit trail. The master registry records every source with its production context and its data-quality assessment. The decision log records every non-trivial judgment, namely inclusions, exclusions, locus assignments, segmentation rules and translation choices, with a date and a rationale. The coverage statement records what was reached and what was not. Where the protocol can be registered or time-stamped in advance, it is, to separate planned decisions from emergent ones. The honest constraint is recorded too, namely that raw community data often cannot be shared openly, so reproducibility is supported through the registry, the decision log and the cleaned, de-identified corpus rather than through release of the raw material (Lamb et al., 2024).

13. Trustworthiness and quality criteria

Rigour is pursued against the four established trustworthiness criteria, each tied to a concrete practice in this build.

Criterion	How it is met here
Credibility	Triangulation of the official record against lived discourse; mapping of every hurdle to representative documents; the data-quality assessment filter
Transferability	Thick description of the production context of sources; the three-verification-logic contrast that shows which hurdles travel

Criterion	How it is met here
Dependability	The decision log and the fixed registry schema, so the process is traceable and consistent
Confirmability	Every topic and hurdle traceable to source rows; machine output separated from human interpretation; the audit trail

14. Saturation and the two-way coverage audit

Growth continues until new sources stop adding new content. Before the build closes, run a two-way audit, first against the four hurdle loci and second against the five verification-logic rows, confirming every cell is populated by at least one source that passed the data-quality assessment. Produce the coverage statement that becomes the basis for the limitations section.

Verification logic \ Locus	Internal	Relational	Institutional	Informational
Spices residue	fill	fill	fill	fill
Marine catch	fill	fill	fill	fill
Value-added process	fill	fill	fill	fill
Rice baseline	fill	fill	fill	fill
Deforestation regime	fill	fill	fill	fill

15. Honest limits to plan around

1. Answerability boundary. The corpus answers the hurdle-landscape question at population and discourse level, not a firm-level causal question.
2. Secondary-data fit. The data was produced for other purposes, so the data-quality assessment and the production-context notes are what keep reuse defensible; sources that fail the screen are excluded and recorded.
3. Asymmetry and access. RASFF is anonymised while FDA refusals are firm-named; pre-2020 RASFF is archived; the IFS database is partly restricted; firm-level shipment data is commercial; X and professional platforms restrict bulk access.
4. GDELT quality. Moderate key-field accuracy and high redundancy require deduplication and cleaning before use.
5. Sharing constraint. Raw community data cannot always be shared openly, so reproducibility rests on the registry, the decision log and the de-identified corpus.

16. Appendix. Exceptions file template

Field	Content
doc_id_attempted source_name	/ Reserved identifier; issuing body or platform

Field	Content
intended_url_or_query / attempt_date	What was attempted; ISO date
reason_inaccessible	paywalled, restricted, archived, broken, terms-of-service, failed-data-quality
workaround_tried / resolution	Archive, alternate source, interface change; resolved, abandoned, deferred
notes	Free text for the limitations section

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